TECHNO INDIA UNIVERSITY
WESTBENGAL

Improved Fungal Growth Optimization (IFGO) and its use in Clustering Improvement

8th Semester Project Report

submitted to

Techno India University, West Bengal

In partial fulfilment of the requirements
for the award of the degree of

Bachelor of Technology

in

Computer Science & Engineering

by

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We hereby declare that the project report entitled **"Improved Fungal Growth Optimization (IFGO) and its use in Clustering Improvement"** is an authentic record of our own work carried out at the **Department of Computer Science and Engineering, Techno India University, West Bengal**, during the **8th semester of the academic year 2025-2026** under the supervision of **Prof. Kalyan Kumar Das**, Assistant Professor.

We further declare that the matter embodied in this project report has not been submitted by us for the award of any other degree or diploma of this or any other Institute/University. All the information have been obtained and presented in accordance with academic rules and ethical conduct. We also declare that, as required by these rules and conduct, we have fully cited and referenced all materials and results that are not original to this work.

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ACKNOWLEDGEMENT

We wish to express our deep appreciation to Prof. Kalyan Kumar Das, our project mentor for his patient guidance, expertise and encouragement throughout this research. His valuable suggestions significantly improved our project.

We are grateful to the Computer Science and Engineering Department faculty for providing the necessary resources and learning environment. Special thanks to our classmate, for their collaboration and insightful discussions.

ABSTRACT

The surge of demands in the development of artificial intelligence has pointed to the importance of finding the optimal cluster centers by clustering algorithms. Fungal Growth Optimization (FGO) is a metaheuristic algorithm that is inspired by the natural expansion, hyphal branching; this can be used to enhance the performance of conventional clustering algorithms. The need for increased efficiency and performance can be achieved by improving this approach. This suggestion has been tested in this work to determine how an Improved Fungal Growth Optimization (IFGO) can improve clustering quality. IFGO changes the position of the cluster centers and improves the cluster in a subsequent iteration after exploration. This result shows that IFGO can be used for complex clustering tasks and can serve as a foundation for hybrid clustering frameworks. Furthermore, the proposed IFGO approach introduces additional mechanisms to improve the balance between exploration and exploitation during the formation of clusters. In the proposed IFGO the clustering process is guided by a population-based search mechanism where multiple candidate solutions do convergent evolution simultaneously. Each candidate solution represents a potential set of cluster centroids and these candidate solutions are refined through fungal expansion and funagl branching behavior.

Improved Fungal Growth Optimization (IFGO) and its use in Clustering Improvement

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Chapter 1

Introduction

Clustering [1] is an unsupervised learning method [2] which is used to cluster similar data points. It reveals hidden structures based on shared datasets. Clustering is important because it does simplify complex data sets, helps to make better decisions and enhances tasks like pattern recognition [3], image segmentation [4] and data compression [5]. Clustering also contributes to the improvement of the performance of analytical models [6] and helps to enable researchers to understand relationships and trends more efficiently.

1.1 Problem Statement

Traditional clustering algorithms [7] do have issues like: it struggles with the initialization of centroid values; it can cause noisy or high-dimensional data if the cluster is non-spherical, and it needs the pre-defined number of clusters, which may lead to a misrepresentation of the true data structure.

1.2 Objectives

To overcome the limitations improvement of clustering is needed, hybrid algorithms are useful to find comparatively better cluster configurations by minimizing objective functions, avoiding local minima and reducing sensitivity to initialization. These algorithms help to find the more accurate cluster boundaries in complex, high-dimensional or, noisy datasets. These leads to the improvement of performance and a more reliable outcome for

a real-world approach. In recent years optimization based hybrid clustering approaches have gained significant attention due to their special ability. These hybrids utilize optimization strategies to search for the best possible cluster configurations.

1.3 Improvements

The proposed IFGO algorithm is the integration of stochastic branching along with nutrient-driven growth modeling. It enables a more realistic simulation of fungal behavior by allowing the algorithm to effectively explore diverse regions of the search space while maintaining convergence toward optimal solutions. Lévy flight-based perturbation is introduced to improve global search capability and to avoid premature convergence. The algorithm includes an elite preservation strategy which ensures that high-quality solutions are retained and guide subsequent iterations. A feedback-based update mechanism is also employed.

Chapter 2

Project Overview

TABLE 2.1: Gantt Chart Breakdown of Project Development

Milestone	Task Description	Dec	Jan	Feb	Mar	Apr
M1	Selection Algorithms	■				
M2	Algorithm Evaluation		■■			
M3	Finalization		■			
M4	Algorithm Development			■■		
M5	Dataset Selection			■		
M6	Result (Graphs & Plots)				■■	
M7	Paper Writing				■	■

Chapter 3

Literature Review

Fungal Growth Optimization (FGO) is a bio inspired optimization algorithm that mimics how fungi grow, spread and search for nutrients in nature. This is chosen as our standard algorithm. For pure FGO algorithm the mathematical formation is given below.

3.1 Mathemetical Expression

3.1.1 Population Initialization

$$X_i = (x_{i1}, x_{i2}, x_{i3}, \ldots, x_{id}) \tag{3.1}$$

$$x_{ij} = x_j^{\min} + rand(0,1) \times (x_j^{\max} - x_j^{\min}) \tag{3.2}$$

3.1.2 Fitness Evaluation

$$f_i = f(X_i) \tag{3.3}$$

3.1.3 Spore Dispersal (Exploration Phase)

$$X_i^{t+1} = X_i^t + r_1 \times (X_{best}^t - X_i^t) \tag{3.4}$$

3.1.4 Mycelium Expansion (Local Exploitation)

$$X_i^{t+1} = X_i^t + r_2 \times (X_j^t - X_k^t) \tag{3.5}$$

3.1.5 Nutrient Absorption

$$X_i^{t+1} = X_i^t + r_3 \times (X_{best}^t - X_{worst}^t) \tag{3.6}$$

3.1.6 Adaptive Growth Factor

$$G(t) = G_0 \times \left(1 - \frac{t}{T}\right) \tag{3.7}$$

3.1.7 Position Update with Growth

$$X_i^{t+1} = X_i^t + G(t) \times rand(0,1) \times (X_{best}^t - X_i^t) \tag{3.8}$$

3.1.8 Boundary Control

$$x_{ij} = \begin{cases} x_j^{\min}, & x_{ij} < x_j^{\min} \\ x_j^{\max}, & x_{ij} > x_j^{\max} \end{cases} \tag{3.9}$$

3.2 Gap Identification

Although clustering has been extensively studied in machine learning and data mining several limitations still exist in traditional and metaheuristic-based clustering approaches. To address these issues there is a need for an improved clustering framework that enhances centroid initialization, strengthens global search capability and provides better robustness across diverse datasets. The proposed Improved Fungal Growth Optimization (IFGO) based clustering approach aims to address these gaps by integrating chaotic initialization, adaptive search strategies, Lévy flight mutation and hybrid clustering mechanisms to achieve more stable clustering performance.

Chapter 4

Related Work

Clustering has been studied in the fields of machine learning and data mining [8] along with traditional algorithms like K-Means [9] and FCM [10] being widely used due to its simplicity and computational efficiency. Traditional clustering methods like K-Means and FCM have some issues like they are highly sensitive to initial parameter settings, they often converge local optima especially for high dimensional datasets. To overcome this issue, metaheuristic algorithms [11] are being integrated. Studies demonstrate that hybrid metaheuristic clustering methods outperform traditional algorithms in terms of robustness, clustering quality and convergence behavior. Integrating different traditional clustering methods with metaheuristic algorithms like Genetic Algorithm (GA) [12], Particle Swarm Optimization (PSO) [13], Gorilla Troop Optimization (GTO) [14], Billiards Optimization (BO) [15], Improved Social Group Optimization (ISGO) [16], Energy Optimization (EO) [17] and Hippo Optimization (HO) [18] can be beneficial to adopt a better clustering method. Several studies have explored the integration of metaheuristic algorithms with clustering techniques in order to improve clustering accuracy and stability. These hybrid approaches attempt to overcome the limitations of traditional clustering methods more effectively. Combining the strengths of metaheuristic optimization with clustering algorithms researchers aim to reduce the risk of premature convergence and improve the overall clustering formation. While gradually refining promising solutions FGO mechanism allows the algorithm to adaptively explore different regions of the search space. Such characteristics make the FGO algorithm suitable for solving clustering problems where identifying optimal cluster centroids is a challenging task. Despite of its promising capabilities the FGO algorithm still encounter certain limitations.

Chapter 5

Methodology

5.1 Algorithm

Algorithm 1 Improved-FGO Based Clustering (IFGO)

Input: Dataset $X = \{x_i\}_{i=1}^{N}$, clusters K, population P, iterations T

Output: Optimal centroids Z^*

Initialize bounds lb, ub from X

Generate logistic chaotic sequence $r_{t+1} = 4r_t(1 - r_t)$

Initialize fungi $Z_{i,j} = lb + r_t(ub - lb),\ i = 1..P$

Evaluate fitness $F(Z_i)$

for $t = 1$ **to** T **do**

- Assign samples to nearest centroid $\ell_i = \arg\min_j \|x_i - c_j\|$
- Compute fitness $F = S + \frac{1}{1+DB} + \frac{CH}{1+CH} + D + \frac{1}{1+XB}$
- Update best solution Z_{best}
- Adaptive parameter $r(t) = r_{\max} - (t/T)^{\beta}(r_{\max} - r_{\min})$
- **for** *each* Z_i **do**
  - Position update $Z_i^{t+1} = Z_i^t + r(t)(Z_{\text{best}}^t - Z_i^t)$
  - Lévy perturbation $Z_i^{t+1} = Z_i^{t+1} + \alpha \cdot Levy(\lambda)$
  - Opposition learning $Z_i^{new} = lb + ub - Z_i$
  - Boundary control
- Local refinement:
- K-Means $c_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i$
- FCM $c_j = \frac{\sum u_{ij}^m x_i}{\sum u_{ij}^m}$

return Z_{best}

Algorithm 1 illustrates the workflow of the proposed IFGO algorithm.

5.2 Mathematical Formulation

5.2.1 Problem Definition

Let the dataset be represented as:

$$X = \{x_1, x_2, \ldots, x_N\}, \quad x_i \in \mathbb{R}^d \tag{5.1}$$

where N denotes the total number of data samples and d represents the feature dimensionality.

The clustering objective is to partition X into K disjoint clusters by determining optimal centroids:

$$C = \{c_1, c_2, \ldots, c_K\}, \quad c_j \in \mathbb{R}^d \tag{5.2}$$

In the proposed Improved-FGO framework, each candidate solution encodes all cluster centroids as:

$$Z_i = [c_1, c_2, \ldots, c_K] \in \mathbb{R}^{K \times d} \tag{5.3}$$

The clustering task is therefore formulated as a constrained optimization problem:

$$\max_{Z} \; F(Z) \tag{5.4}$$

subject to:

$$c_j \in [lb, ub]^d \tag{5.5}$$

where lb and ub denote lower and upper bounds derived from the dataset feature space.

5.2.2 Cluster Assignment Mechanism

For a given solution Z, each data sample is assigned to its nearest centroid using Euclidean distance formula:

$$\ell_i = \arg \min_{j \in \{1,\ldots,K\}} \|x_i - c_j\|_2 \tag{5.6}$$

This assignment produces K clusters $C_1, C_2, \ldots, C_K$.

5.2.3 Composite Fitness Function

To enhance clustering robustness and structural validation, a multi-objective fitness function is constructed by integrating five validity indices:

$$F = S + \frac{1}{1+DB} + \frac{CH}{1+CH} + D + \frac{1}{1+XB} \tag{5.7}$$

Here, DB and XB are transformed into maximization form.

For intermediate iterations, a computationally efficient proxy objective is employed:

$$F = -\sum_{i=1}^{N} \min_{j} \|x_i - c_j\| \tag{5.8}$$

5.2.4 Silhouette Index

$$S = \frac{1}{N} \sum_{i=1}^{N} \frac{b(i) - a(i)}{\max(a(i),\, b(i))} \tag{5.9}$$

where $a(i)$ denotes the intra-cluster distance and $b(i)$ represents the nearest-cluster distance.

5.2.5 Davies–Bouldin Index

$$DB = \frac{1}{K} \sum_{i=1}^{K} \max_{j \neq i} \left(\frac{\sigma_i + \sigma_j}{\|c_i - c_j\|} \right) \tag{5.10}$$

where σ_i is the average intra-cluster dispersion of cluster i.

5.2.6 Calinski–Harabasz Index

$$CH = \frac{\mathrm{Tr}(B_K)/(K-1)}{\mathrm{Tr}(W_K)/(N-K)} \tag{5.11}$$

where B_K and W_K represent the between-cluster and within-cluster scatter matrices, respectively.

5.2.7 Dunn Index

$$D = \frac{\min_{i \neq j} \delta(C_i, C_j)}{\max_k \Delta(C_k)} \tag{5.12}$$

where $\delta(C_i, C_j)$ denotes the inter-cluster distance and $\Delta(C_k)$ represents the cluster diameter.

5.2.8 Xie–Beni Index

$$XB = \frac{\sum_{j=1}^{K} \sum_{x_i \in C_j} \|x_i - c_j\|^2}{N \cdot \min_{i \neq j} \|c_i - c_j\|^2} \tag{5.13}$$

5.2.9 Chaotic Initialization

Population diversity is enhanced using logistic chaotic mapping:

$$r_{t+1} = 4r_t(1 - r_t) \tag{5.14}$$

Initial positions are generated as:

$$Z_{i,j} = lb + r_t(ub - lb) \tag{5.15}$$

5.2.10 Adaptive Position Update Strategy

To balance exploration and exploitation, an adaptive coefficient is defined as:

$$r(t) = r_{\max} - \left(\frac{t}{T}\right)^{\beta} (r_{\max} - r_{\min}) \tag{5.16}$$

where T denotes the maximum number of iterations.

The fungi position update rule is expressed as:

$$Z_i^{(t+1)} = Z_i^t + r(t)\left(Z_{\text{best}}^t - Z_i^t\right) \tag{5.17}$$

5.2.11 Lévy Flight Exploration

To prevent premature convergence, Lévy perturbation is introduced:

$$Z_i^{(t+1)} = Z_i^{(t+1)} + \alpha \cdot \text{Levy}(\lambda) \tag{5.18}$$

5.2.12 Opposition-Based Learning

For poorly performing solutions, opposition learning is applied:

$$Z_i^{\text{new}} = lb + ub - Z_i \tag{5.19}$$

5.2.13 K-Means Update

$$c_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i \tag{5.20}$$

5.2.14 Fuzzy C-Means Update

Membership:

$$u_{ij} = \frac{\left(\frac{1}{\|x_i - c_j\|}\right)^{\frac{2}{m-1}}}{\sum_{k=1}^{K} \left(\frac{1}{\|x_i - c_k\|}\right)^{\frac{2}{m-1}}} \tag{5.21}$$

Centroid update:

$$c_j = \frac{\sum_{i=1}^{N} u_{ij}^m x_i}{\sum_{i=1}^{N} u_{ij}^m} \tag{5.22}$$

5.2.15 Final Optimization Model

The overall IFGO based clustering model is defined as:

$$\max_{Z \in [lb,\, ub]^{K \times d}} F(Z) \tag{5.23}$$

5.3 Flow Chart

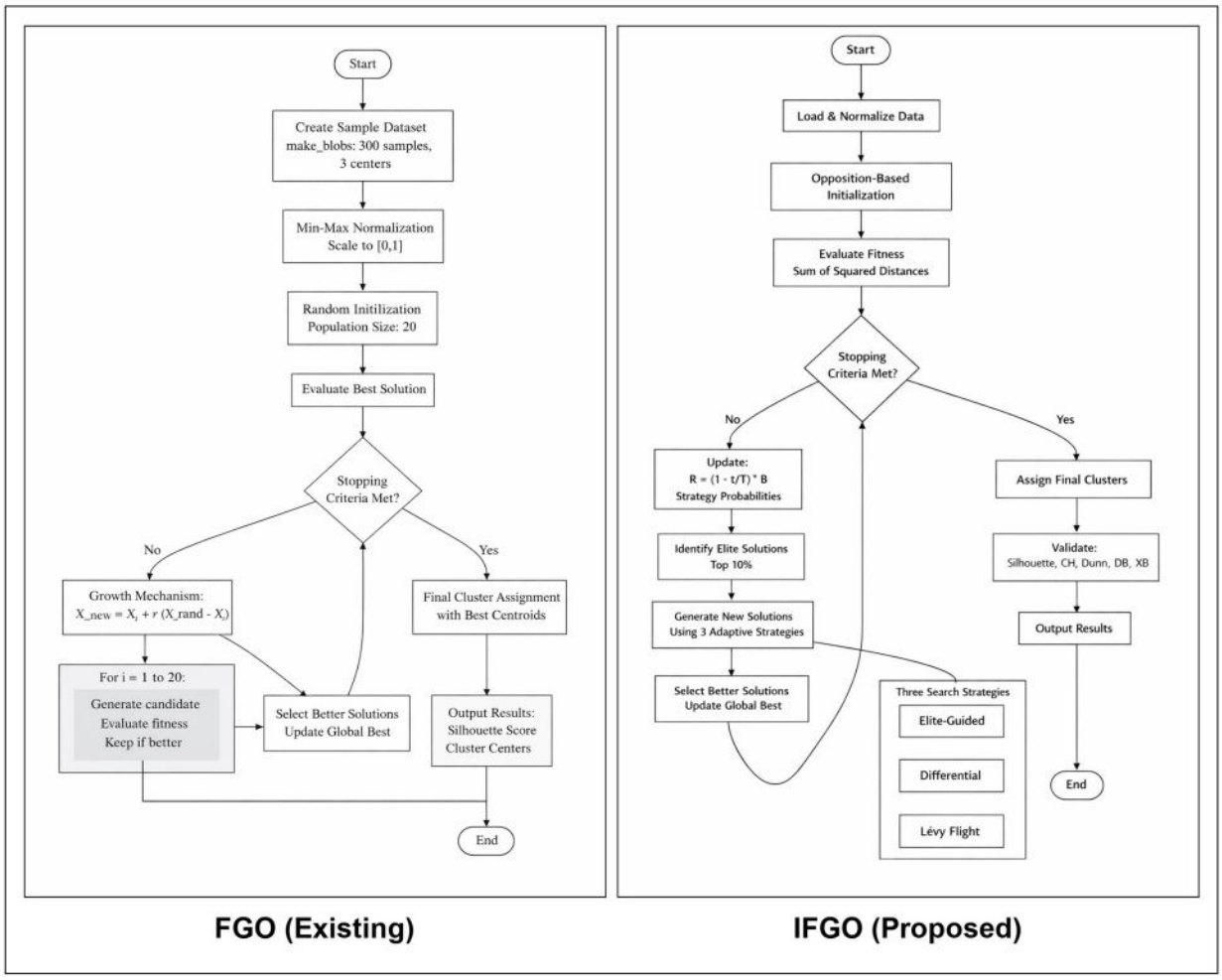


FIGURE 5.1: Comparative Flowchart of FGO and IFGO Algorithms

Figure 5.1 displays a comparative flowchart of the traditional FGO and the proposed algorithm. It highlights their initialization, iterative improvement process, clustering steps and final result generation. IFGO contains structural enhancements and better solution-updating mechanisms.

TABLE 5.1: Comparison between FGO and IFGO

Aspects	**FGO (Existing)**	**IFGO (Proposed)**
Mutation Strategy	No mutation operator	Lévy flight based mutation
Population Initialization	Random initialization	Opposition-based learning (OBL)
Initialization Formula	$X_i \sim U(lb, ub)$	$X_{\text{opposite}} = lb + ub - X$
Exploration Mechanism	Random interaction	Differential exploration using $(X_j - X_k)$ vectors
Strategy Adaptation	Static search behaviour	Adaptive strategy selection based on success probability

Table 5.1 showcases the difference between the FGO and the proposed IFGO algorithm. It highlights the key modifications introduced in the proposed IFGO.

Chapter 6

Novel Contributions

6.1 Novelty

The proposed Improved Fungal Growth Optimization (IFGO) algorithm introduces a structurally enhanced and self-adaptive optimization framework for clustering. Unlike conventional metaheuristic modifications that rely on isolated improvements, IFGO integrates multiple mechanisms within a unified adaptive search paradigm. The key novel contributions are summarized as follows:

1. **Self-Adaptive Multi-Strategy Optimization Mechanism:**

   A dynamic strategy selection framework is introduced in which multiple search operators—elite-guided exploitation, differential exploration, and Lévy-based mutation—compete during the optimization process. Instead of using fixed probabilities, IFGO employs a success-based adaptation mechanism that continuously updates the selection probability of each strategy according to its historical performance. This enables automatic prioritization of effective search behaviors, suppression of underperforming operators, and data-driven adaptation of the search process, transforming the optimizer into a self-learning system.

2. **Elite-Mean Guided Search for Stable Convergence:**

   Unlike other traditional metaheuristics that are greatly affected by the best global solution, IFGO uses the elite mean guidance strategy to determine its search direction based on the mean of the best-performing solutions. This minimizes the effects of noise-affected solutions and maximizes the diversity of solutions.

3. **Strategy-Level Adaptive Exploration–Exploitation Control:**

   IFGO strikes a balance between exploration and exploitation through the adaptive switching of search behaviors. Elite-guided updates facilitate exploitation, differential interactions facilitate exploration, and Lévy flights facilitate jumps. The switching of search behaviors enables the adaptation of the optimization process based on the state of the search.

4. **Closed-Loop Feedback-Driven Optimization:**

   A feedback loop between the quality of solutions and the behavior of the search process is established, where improvements in fitness directly affect the selection of future strategies. This creates a form of reinforcement learning, which continually enhances the adaptability of the search process in complex optimization landscapes.

5. **Diversity-Preserving Initialization with Opposition-Based Sampling:**

   IFGO also includes a better initialization approach that involves a combination of random population generation with opposition-based solutions. This approach for sampling the population space is likely to improve the chances of exploring better areas in the space during the initial iterations.

6. **Composite Multi-Objective Fitness Formulation for Clustering:**

   A single fitness function is constructed using a combination of different cluster validity measures, i.e., Silhouette, Davies-Bouldin, Calinski-Harabasz, Dunn, and Xie-Beni. This helps in optimizing all aspects of cluster compactness, separation, and integrity.

7. **Hybrid Optimization–Refinement Framework:**

   The proposed IFGO is combined with K-Means and Fuzzy C-Means (FCM) algorithms, and a new hybrid model is created. The global optimization for the position of centroids is performed by the proposed IFGO, and local search is performed by using the K-Means and Fuzzy C-Means algorithms.

Overall, the proposed IFGO framework introduces a self-adaptive, feedback-driven, multi-strategy optimization paradigm that significantly enhances clustering performance, convergence stability, and robustness across diverse datasets.

TABLE 6.1: Comparison of Optimization Features Across Algorithms

Algorithm	**Lévy**	**OBL**	**Adaptive**	**Multi-objective**	**Hybrid**	**Self-adaptive**
FGO	×	×	×	×	×	×
PSO	×	×	Partial	×	✓	×
Proposed	✓	✓	✓	✓	✓	✓

6.2 Computational Complexity

The computational complexity of the proposed Improved Fungal Growth Optimization (IFGO) algorithm depends on the population size P, number of iterations T, dataset size N, number of clusters K, and feature dimension d.

In each iteration, the dominant computational cost arises from the cluster assignment process, where each data point is assigned to its nearest centroid. This requires computing distances between N data points and K centroids, resulting in a complexity of $\mathcal{O}(NK)$ per candidate solution. Considering P candidate solutions over T iterations, the overall time complexity of the algorithm can be expressed as:

$$\mathcal{O}(PTNK) \tag{6.1}$$

Additional operations such as Lévy flight-based mutation and opposition-based learning introduce an extra computational cost of $\mathcal{O}(P)$ per iteration. However, this cost is relatively negligible compared to the dominant clustering computation.

In comparison, traditional clustering algorithms such as K-Means and Fuzzy C-Means (FCM) exhibit a time complexity of $\mathcal{O}(NKT)$. The higher computational cost of IFGO is attributed to its population-based optimization framework. Despite this, IFGO provides improved global search capability, better avoidance of local optima, and enhanced clustering performance.

The space complexity of the proposed algorithm is determined by the storage of candidate solutions and is given by (where d denotes the dimensionality of the dataset):

$$\mathcal{O}(PKd) \tag{6.2}$$

Chapter 7

Results and Discussion

The results are presented here. The following cluster validity indices and statistical analysis are used to compare the quality of over all formed clusters.

Silhouette Score [19] measures the degree to which each data point fits within its assigned cluster. High value represent better separation, cohesion between the formed clusters.

Davies-Bouldin Index [20] evaluates the mean similarity between clusters. Less value represent clustering with more distinct groups.

Dunn Index [21] evaluates cluster separation and compactness. High value indicate well separated clusters.

Xie-Beni Index [22] assesses compactness and separation. Low value represent well defined clusters.

Caliński-Harabasz Score [23] is used to determine the optimal number of clusters. Higher value shows better results.

TABLE 7.1: Description of the Datasets used in the Experimental Study

Dataset	**Domain**	**Sample**	**Features**
Iris Dataset	Botany	150	6
Drug Dataset	Healthcare / Pharmacology	1,000	9
Bank Dataset	Finance	2,512	16
Shopping Dataset	E-Commerce	3,900	18
Wave Dataset	Signal Processing	12,762	3
Placement Dataset	Education	10,000	10
Delivery Dataset	Supply Chain	25,000	15

Table 7.1 summarizes the domain, number of samples, and features count for each dataset used in the study. These datasets are available in the public domain, downloaded from the website Kaggle [24].

7.1 Comparison of Clustering Validity Indices

TABLE 7.2: Comparison of Clustering Validity Indices on the Bank Dataset

Algorithm	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
IFGO+K-Means	0.280317	0.318055	919.974371	0.022518	1.428218
IFGO+FCM	0.276693	0.535317	911.964690	0.022518	1.456834
FGO+K-Means	0.250692	0.784791	723.976616	0.019010	1.323301
FGO+FCM	0.200912	3.822037	488.605095	0.009445	1.900755
K-Means	0.250600	0.784498	723.959000	0.019000	1.323900
FCM	0.200800	3.908100	488.709500	0.009400	1.900743
GA+K-Means	0.250731	0.785700	723.974949	0.019010	1.323019
GA+FCM	0.200803	3.869657	488.509370	0.009445	1.899800
PSO+K-Means	0.250692	0.784791	723.976616	0.019010	1.323301
PSO+FCM	0.200923	3.820879	488.669571	0.009445	1.900030
GTO+K-Means	0.250692	0.784791	723.976616	0.019010	1.323301
GTO+FCM	0.200864	3.881321	488.437263	0.009445	1.901460
BO+K-Means	0.250692	0.784791	723.976616	0.019010	1.323301
BO+FCM	0.200803	3.850107	488.509370	0.009445	1.900743
ISGO+K-Means	0.250692	0.784791	723.976616	0.019010	1.323301
ISGO+FCM	0.200864	3.880952	488.437263	0.009445	1.901460
EO+K-Means	0.250692	0.784791	723.976616	0.019010	1.323301
EO+FCM	0.200864	3.880117	488.437263	0.009445	1.901460
HO+K-Means	0.250692	0.784791	723.976616	0.019010	1.323301
HO+FCM	0.200991	3.821126	488.649606	0.009445	1.900537

Table 7.2 showcases the comparison of clustering validity indices on the Bank Dataset. For Bank Dataset, the IFGO integrated with K-Means outperforms other K-Means based hybrid algorithms by producing highest Silhouette Score, CH Score, Dunn Index and lowest XB index.

TABLE 7.3: Comparison of Clustering Validity Indices on the Delivery Dataset

Algorithm	**Silhouette**	**XB-Index**	**CH-Score**	**Dunn-Index**	**DB-Index**
IFGO+K-Means	0.216182	0.489173	7841.937318	0.007251	1.652433
IFGO+FCM	0.139232	0.941437	6782.355194	0.004275	2.247269
FGO+K-Means	0.221523	0.860723	8132.152518	0.009185	1.627833
FGO+FCM	0.141532	1.433497	6931.204013	0.006222	2.215477
K-Means	0.221500	0.860600	8132.150400	0.009200	1.628000
FCM	0.141600	1.440400	6931.580100	0.006200	2.215500
GA+K-Means	0.221523	0.860723	8132.152518	0.009185	1.627833
GA+FCM	0.141540	1.433522	6931.208537	0.006222	2.215711
PSO+K-Means	0.221523	0.860723	8132.152518	0.009185	1.627833
PSO+FCM	0.141548	1.433481	6931.114820	0.006222	2.215711
GTO+K-Means	0.221523	0.860723	8132.152518	0.009185	1.627833
GTO+FCM	0.141531	1.433343	6931.141342	0.006222	2.215685
BO+K-Means	0.221523	0.860723	8132.152518	0.009185	1.627833
BO+FCM	0.141532	1.433507	6931.204013	0.006222	2.215477
ISGO+K-Means	0.221523	0.860723	8132.152518	0.009185	1.627833
ISGO+FCM	0.141531	1.433552	6931.141342	0.006222	2.215685
EO+K-Means	0.221523	0.860723	8132.152518	0.009185	1.627833
EO+FCM	0.141547	1.433487	6931.080421	0.006222	2.215774
HO+K-Means	0.221523	0.860723	8132.152518	0.009185	1.627833
HO+FCM	0.141525	1.433530	6931.147928	0.006222	2.215446

Table 7.3 shows that for Delivery Dataset the K-Means based hybrid algorithms outperforms FCM based hybrid algorithms. Metaheuristic optimization does not make a meaningful improvement; K-Means provides the optimal solution.

TABLE 7.4: Comparison of Clustering Validity Indices on the Drug Dataset

Algorithm	**S-Score**	**XB-Index**	**CH-Index**	**Dunn-Index**	**DB-Index**
IFGO+K-Means	0.248219	0.360245	391.071300	0.027001	1.394292
IFGO+FCM	0.250264	0.493597	388.715045	0.032327	1.389608
FGO+K-Means	0.212309	0.780998	255.813186	0.040165	1.600545
FGO+FCM	0.203297	0.963209	239.140871	0.026304	1.578206
K-Means	0.212300	0.781000	255.813200	0.040200	1.600500
FCM	0.202500	0.967000	238.309000	0.026300	1.577400
GA+K-Means	0.212309	0.780998	255.813186	0.040165	1.600545
GA+FCM	0.203297	0.963212	239.140871	0.026304	1.578206
PSO+K-Means	0.212309	0.780998	255.813186	0.040165	1.600545
PSO+FCM	0.203297	0.963209	239.140871	0.026304	1.578206
GTO+K-Means	0.212309	0.780998	255.813186	0.040165	1.600545
GTO+FCM	0.203297	0.963209	239.140871	0.026304	1.578206
BO+K-Means	0.212309	0.780998	255.813186	0.040165	1.600545
BO+FCM	0.203297	0.962145	239.140871	0.026304	1.578206
ISGO+K-Means	0.212309	0.780998	255.813186	0.040165	1.600545
ISGO+FCM	0.203297	0.963212	239.140871	0.026304	1.578206
EO+K-Means	0.212309	0.780998	255.813186	0.040165	1.600545
EO+FCM	0.203297	0.963209	239.140871	0.026304	1.578206
HO+K-Means	0.212309	0.780998	255.813186	0.040165	1.600545
HO+FCM	0.203297	0.963209	239.140871	0.026304	1.578206

Table 7.4 showcases that hybrid algorithms based on IFGO outperforms all other algorithms. IFGO significantly enhances clustering quality over all for Drug Dataset. IFGO shows superior cluster separation and compactness with stronger variance-based structure.

TABLE 7.5: Comparison of Clustering Validity Indices on the Iris Dataset

Algorithm	**Silhouette**	**XB-Index**	**CH-Score**	**Dunn-Index**	**DB-Index**
IFGO+K-Means	0.520362	0.110525	365.452968	0.133525	0.765732
IFGO+FCM	0.520362	0.114563	365.452968	0.133525	0.765732
FGO+K-Means	0.452950	0.304155	237.842018	0.082819	0.896584
FGO+FCM	0.452319	0.230330	237.673998	0.082819	0.895600
K-Means	0.452900	0.304200	237.842000	0.082800	0.896600
FCM	0.452300	0.230600	237.674000	0.082800	0.895600
GA+K-Means	0.452950	0.304155	237.842018	0.082819	0.896584
GA+FCM	0.452319	0.230600	237.673998	0.082819	0.896584
PSO+K-Means	0.452950	0.304155	237.842018	0.082819	0.896584
PSO+FCM	0.452319	0.230308	237.673998	0.082819	0.895600
GTO+K-Means	0.452950	0.304155	237.842018	0.082819	0.896584
GTO+FCM	0.452319	0.230335	237.673998	0.082819	0.895600
BO+K-Means	0.452950	0.304155	237.842018	0.082819	0.896584
BO+FCM	0.452319	0.230343	237.673998	0.082819	0.895600
ISGO+K-Means	0.452950	0.304155	237.842018	0.082819	0.896584
ISGO+FCM	0.452319	0.230408	237.673998	0.082819	0.895600
EO+K-Means	0.452950	0.304155	237.842018	0.082819	0.896584
EO+FCM	0.452319	0.230421	237.673998	0.082819	0.895600
HO+K-Means	0.452950	0.304155	237.842018	0.082819	0.896584
HO+FCM	0.452319	0.230419	237.673998	0.082819	0.895600

Table 7.5 shows that for Iris Dataset IFGO based hybrid algorithm provides the highest Silhouette Score, CH score, Dunn index along with lowest XB index and DB index. Other metaheuristic based hybrid algorithms show nearly identical values of clustering validity indices.

TABLE 7.6: Comparison of Clustering Validity Indices on the Placement Dataset

Algorithm	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
IFGO+K-Means	0.139580	0.754380	1726.659193	0.027293	2.072543
IFGO+FCM	0.172075	13879.100830	2241.978226	0.029667	2.065217
FGO+K-Means	0.143576	1.551743	1873.041327	0.016483	2.160249
FGO+FCM	0.060793	262.153398	1465.828933	0.029239	3.430847
K-Means	0.143500	1.551900	1873.035100	0.016500	2.160249
FCM	0.061000	264.199400	1465.967300	0.029200	3.431200
GA+K-Means	0.143576	1.551743	1873.041327	0.016483	2.160249
GA+FCM	0.060958	264.187475	1465.967348	0.029239	3.431195
PSO+K-Means	0.143576	1.551743	1873.041327	0.016483	2.160249
PSO+FCM	0.060793	262.091415	1465.828933	0.029239	3.430847
GTO+K-Means	0.143576	1.551743	1873.041327	0.016483	2.160249
GTO+FCM	0.060958	264.232995	1465.967348	0.029239	3.431195
BO+K-Means	0.143576	1.551743	1873.041327	0.016483	2.160249
BO+FCM	0.060958	264.491498	1465.967348	0.029239	3.431195
ISGO+K-Means	0.143576	1.551743	1873.041327	0.016483	2.160249
ISGO+FCM	0.060958	264.359163	1465.967348	0.029239	3.431195
EO+K-Means	0.143576	1.551743	1873.041327	0.016483	2.160249
EO+FCM	0.060958	264.232699	1465.967348	0.029239	3.431195
HO+K-Means	0.143576	1.551743	1873.041327	0.016483	2.160249
HO+FCM	0.060957	264.823884	1465.804091	0.029239	3.431841

Table 7.6 showcases nearly identical low Silhouette Score across all hybrid algorithms; informing that the clustering structure in Placement Dataset appears weak. Though IFGO moderately enhances K-Means and FCM performances. The extreme XB index value suggests possible instability in the Placement Dataset.

TABLE 7.7: Comparison of Clustering Validity Indices on the Shopping Dataset

Algorithm	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
IFGO+K-Means	0.139580	0.754380	1726.659193	0.027293	2.072543
IFGO+FCM	0.082093	10876.784630	440.794801	0.021867	2.179251
FGO+K-Means	0.138171	1.033940	635.142676	0.038299	1.971172
FGO+FCM	0.130630	283058.983000	596.841064	0.032192	2.031137
K-Means	0.138300	1.042200	634.735800	0.022800	1.976400
FCM	0.126600	619807.377500	582.165300	0.026200	2.051000
GA+K-Means	0.138171	1.033940	635.142676	0.038299	1.971172
GA+FCM	0.130828	315311.005800	594.080995	0.038622	2.044415
PSO+K-Means	0.138171	1.033940	635.142670	0.038299	1.971172
PSO+FCM	0.130404	263630.893600	594.344436	0.027982	2.029268
GTO+K-Means	0.138171	1.033940	635.142676	0.038299	1.971172
GTO+FCM	0.130790	261871.575400	591.964607	0.035306	2.040989
BO+K-Means	0.138171	1.033940	635.142676	0.038299	1.971172
BO+FCM	0.130809	272393.763800	594.784106	0.025731	2.040332
ISGO+K-Means	0.138171	1.033940	635.142676	0.038299	1.971172
ISGO+FCM	0.131314	284890.585200	594.239679	0.021474	2.033080
EO+K-Means	0.138171	1.033940	635.142676	0.038299	1.971172
EO+FCM	0.131560	276193.941300	598.118716	0.037344	2.035323
HO+K-Means	0.138171	1.033940	635.142676	0.038299	1.971172
HO+FCM	0.131437	260650.558000	596.383926	0.028656	2.038899

Table 7.7 shows for Shopping Dataset FCM based hybrid algorithms demonstrate instability indicating poor compactness and potential overlap. Among all hybrid algorithms IFGO+K-Means provides the most balanced performance for Shopping Dataset by improving CH score and XB indices.

TABLE 7.8: Comparison of Clustering Validity Indices on the Wave Dataset

Algorithm	**Silhouette**	**XB-Index**	**CH-Score**	**Dunn-Index**	**DB-Index**
IFGO+K-Means	0.531432	0.114456	17782.387210	0.005234	0.735842
IFGO+FCM	0.525469	0.131566	17585.589140	0.002503	0.737369
FGO+K-Means	0.463731	0.305390	11012.343050	0.002605	0.904901
FGO+FCM	0.462599	0.269499	11002.592070	0.001215	0.903548
K-Means	0.463700	0.305400	11012.343000	0.002600	0.904900
FCM	0.462600	0.269500	11002.592100	0.001200	0.903500
GA+K-Means	0.463731	0.305390	11012.343050	0.002605	0.904901
GA+FCM	0.462587	0.269504	11002.385020	0.001215	0.903535
PSO+K-Means	0.463731	0.305390	11012.343050	0.002605	0.904901
PSO+FCM	0.462587	0.269498	11002.385020	0.001215	0.903535
GTO+K-Means	0.463731	0.305390	11012.343050	0.002605	0.904901
GTO+FCM	0.462587	0.269503	11002.385020	0.001215	0.903535
BO+K-Means	0.463731	0.305390	11012.343050	0.002605	0.904901
BO+FCM	0.462587	0.269501	11002.385020	0.001215	0.903535
ISGO+K-Means	0.463731	0.305390	11012.343050	0.002605	0.904901
ISGO+FCM	0.462599	0.269499	11002.592070	0.001215	0.903548
EO+K-Means	0.463731	0.305390	11012.343050	0.002605	0.904901
EO+FCM	0.462599	0.269498	11002.592070	0.001215	0.903548
HO+K-Means	0.463731	0.305390	11012.343050	0.002605	0.904901
HO+FCM	0.462587	0.269499	11002.385020	0.001215	0.903535

Table 7.8 shows that the proposed IFGO significantly enhances clustering performance in Wave Datasets with strong intrinsic structure. An improvement is observed across all clustering validity indices compared to conventional metaheuristic based hybrids algorithms.

7.2 Statistical Analysis

7.2.1 Boxplot

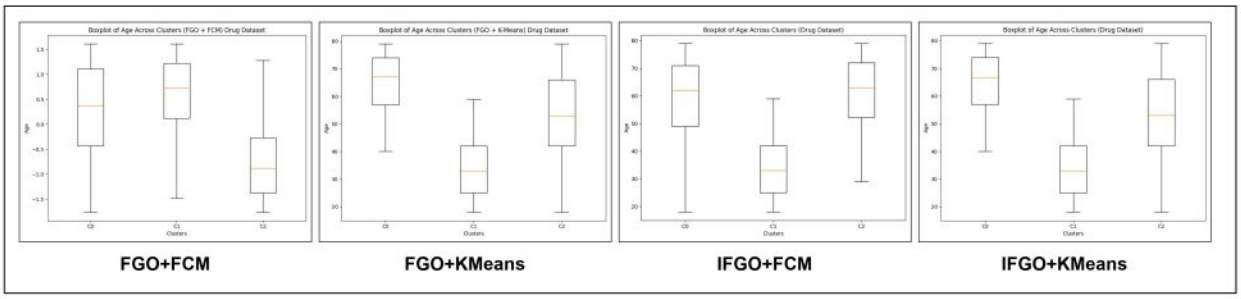


FIGURE 7.1: Boxplot for the Drug Dataset

Figure 7.1 demonstrates that in IFGO variants the median lines of clusters are more distinctly separated compared to the FGO variants. The median age values for FGO+KMeans are C0 = 67, C1 = 33 and C2 = 53, for IFGO+KMeans values are C0 = 66, C1 = 32 and C2 = 54. These values show slightly improved separation for the K-Means variants. On the other hand for FGO+FCM the normalized medians are C0 = 0.35, C1 = 0.70 and C2 = -0.90, whereas IFGO+FCM reflects clearer spacing with C0 = 62, C1 = 33 and C2 = 63. These cause less overlap between clusters, better discrimination between age groups and stronger structure in the Drug Dataset. Furthermore, the reduced inter-quartile range and minimal overlap observed in IFGO based clusters indicate lower intra-cluster variability and improved consistency which reinforces the statistical robustness of the clustering results.

7.2.2 Coefficient of Variation

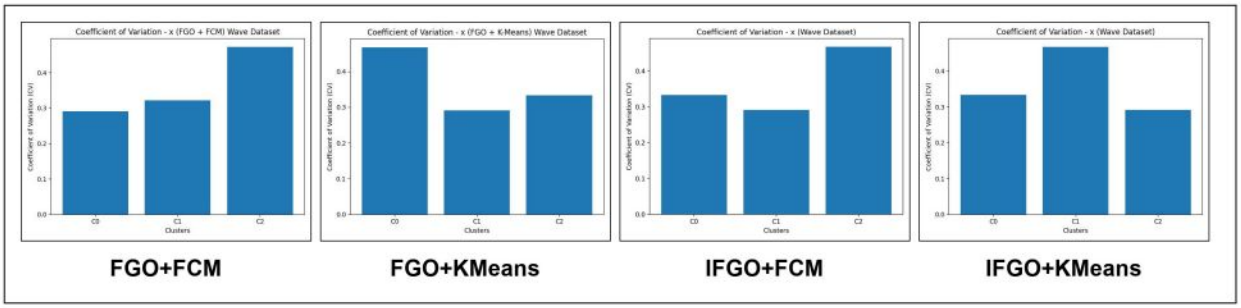


FIGURE 7.2: Coefficient of Variation Analysis for the Wave Dataset

Figure 7.2 shows that for the Wave Dataset IFGO gives more consistent and balanced clustering. IFGO variants show more balanced CV distribution across clusters and on the other hand FGO shows larger fluctuations between clusters. IFGO-based hybrid algorithms provide the best exploration–exploitation balance and centroid positioning stability. As reflected by the values, where FGO+FCM shows C0 = 0.29, C1 = 0.32 and C2 = 0.47 indicating higher imbalance, while FGO+KMeans presents C0 = 0.46, C1 = 0.29 and C2 = 0.34 with noticeable fluctuations. In contrast, IFGO+FCM exhibits relatively stable values of C0 = 0.33, C1 = 0.29 and C2 = 0.46 and IFGO+KMeans further improves consistency with C0 = 0.33, C1 = 0.46 and C2 = 0.29 demonstrating more uniform distribution of variability across clusters. These hybrids are also good at avoiding the local optima.

7.2.3 Convergence Graph

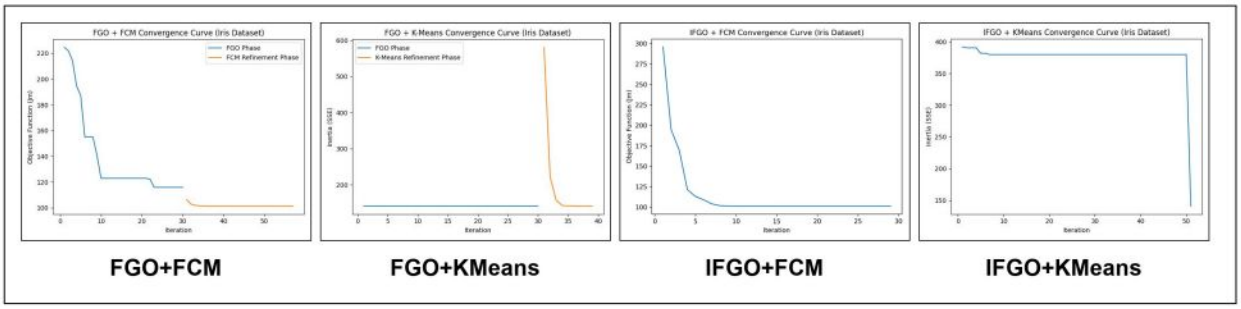


FIGURE 7.3: Convergence Graph for the Iris Dataset

Figure 7.3 displays that IFGO variants provide faster convergence which results with better optimization efficiency. For the Iris Dataset IFGO-based hybrids reach equal or better minima with more stability. IFGO-based hybrids provide better initial cluster centres that reduces burden on the refinement phase. As observed FGO+FCM reduces from 230 to 100 gradually while FGO+KMeans remains nearly stagnant. IFGO+FCM converges rapidly from 295 to 100 within a few iterations and IFGO+KMeans shows stable behavior. This indicates faster stabilization and more efficient optimization in IFGO variants. These algorithms also avoid premature stagnation and achieve competitive or better objective values.

7.2.4 Friedman Test

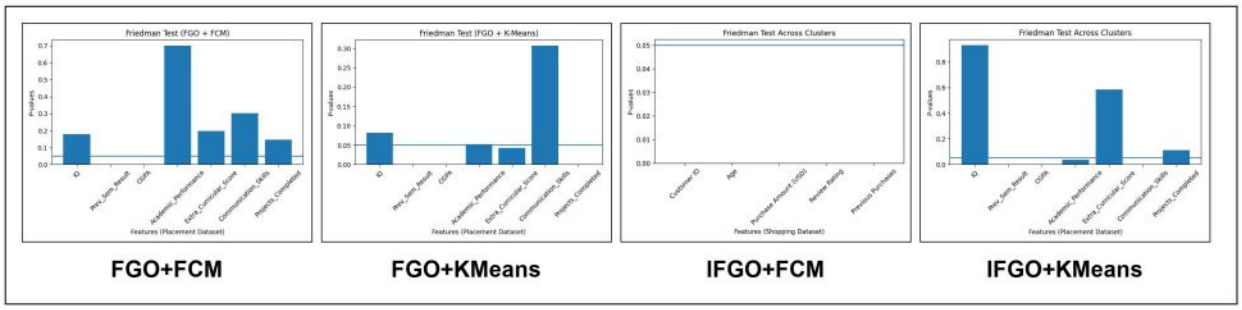


FIGURE 7.4: Friedman Test for the Placement Dataset

Figure 7.4 reveals IFGO+FCM provides p-values tightly aligned around the significance threshold and more consistent statistical behaviour across features. FGO+FCM shows higher variability (e.g. CGPA = 0.70 ; Communication Skills = 0.30) and FGO+KMeans also exhibits fluctuations (e.g. Projects Completed = 0.30). IFGO+KMeans produces lower p-values for key features such as Academic Performance = 0.05 and Projects Completed = 0.12, indicating improved statistical significance. For the Placement Dataset IFGO variants provides clusters that are statistically distinguishable and ranking consistency along with distinct differences between clusters.

7.2.5 T-Test

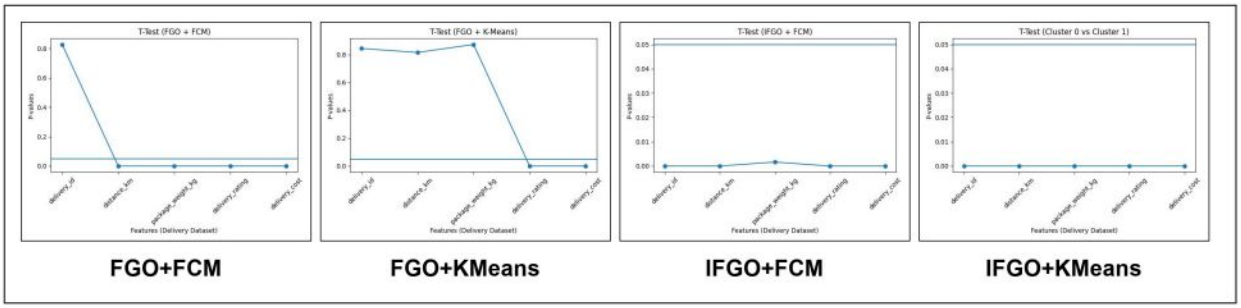


FIGURE 7.5: T-Test for the Delivery Dataset

Figure 7.5 presents that compared to FGO-based algorithms IFGO variants produces stronger mean differences between clusters, statistically significant separation across all features and no weak or inconsistent dimensions along with more reliable segmentation. IFGO is better because it creates statistically significant cluster differences, eliminates weak feature separation and ensures consistent discrimination across dimensions. FGO+FCM shows high values for features like delivery_id = 0.82 indicating no significant difference. FGO+KMeans exhibits weak significance for distance_km = 0.81 and package_weight_kg = 0.87. Whereas IFGO variants produce near zero p-values across all features. This indicates strong statistical significance and consistent cluster differentiation.

7.2.6 Wilcoxon Signed-Rank Test

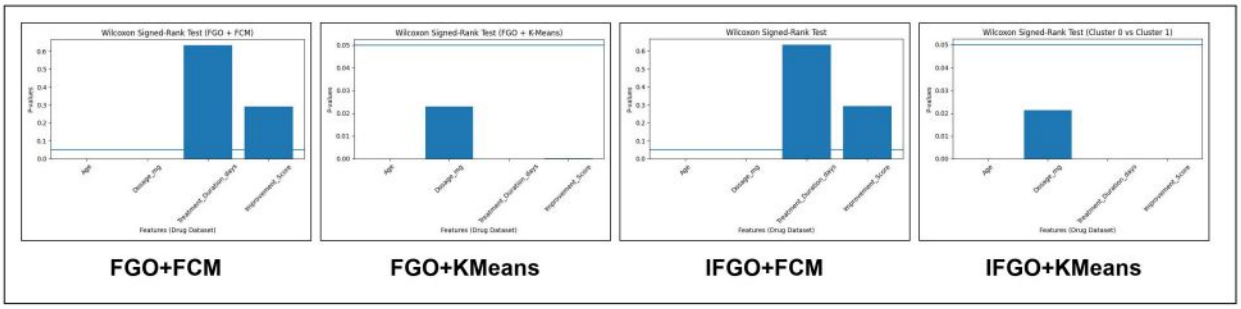


FIGURE 7.6: Wilcoxon Signed-Rank Test for the Drug Dataset

Figure 7.6 suggests that IFGO variants provide superior statistical significance with lower p-values. IFGO is more sensitive to the actual variables in the Drug Dataset. FGO+FCM shows higher values for Treatment Duration = 0.63 and Improvement Score = 0.29, it indicates weaker significance. IFGO variants produce consistently lower p-values (e.g. Dosage = 0.02 and near 0.00 for some features), it confirms stronger statistical reliability. It provides a higher confidence level for the final model, making it a safer choice for clinical application.

7.2.7 IQR Plot

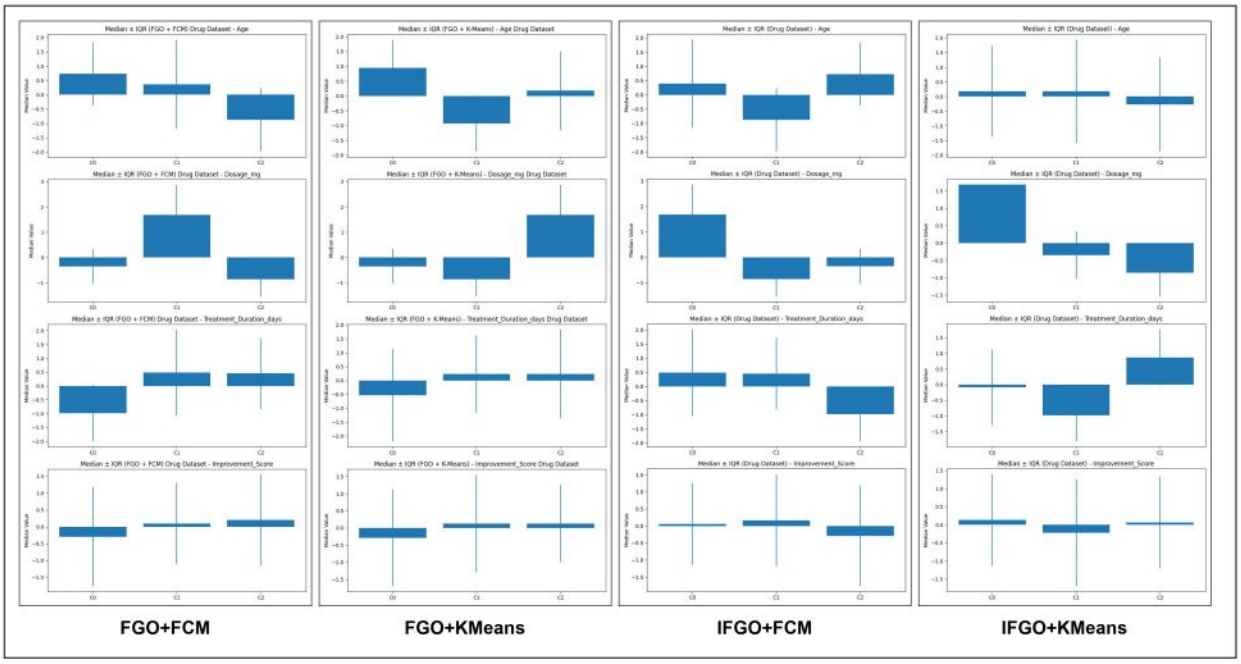


FIGURE 7.7: IQR Plot for the Drug Dataset

Figure 7.7 indicates that for FGO-based algorithms several medians are very close, some clusters overlap heavily and cluster distinction is not sharp for some features. On the other hand IFGO variants produce more distinctly spaced medians, a clearer identity cluster and less ambiguity between C0, C1, and C2. The figure shows that for features like Dosage (mg) and Treatment Duration (days) FGO variants show overlapping median ranges (e.g. C1 and C2 around 0 to 0.5) while IFGO variants exhibit clearer separation (e.g. C0 = 1.5, C1 = -0.5, C2 = -0.2) along with tighter IQRs. Clear median separation means better cluster interpret-ability which gives IFGO a better acceptability to use.

7.2.8 Nemenyi Post-hoc Test

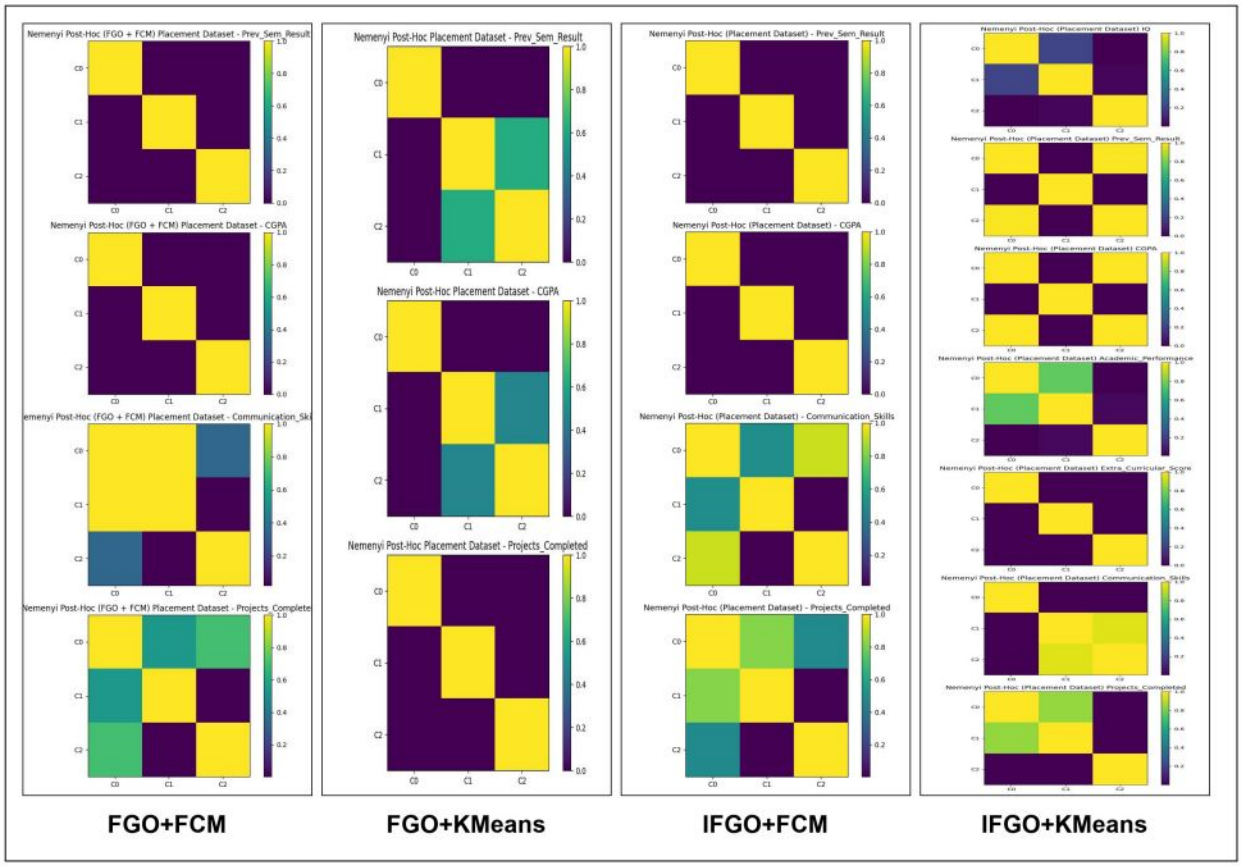


FIGURE 7.8: Nemenyi Post-hoc Test for the Placement Dataset

Figure 7.8 exhibits that IFGO+FCM shows more distinct low p-value blocks, stronger cluster-to-cluster discrimination and fewer ambiguous comparisons. IFGO+KMeans shows most consistent and clean separation, clear statistical difference between clusters and very structured pairwise significance pattern. FGO variants exhibit several mid-range p-values (0.3–0.7) indicating ambiguity between clusters. IFGO variants show more extreme values with many comparisons close to 0 and diagonals near 1. These results confirm clearer and more reliable separation across cluster pairs. Overall IFGO is a better performer.

7.2.9 Vargha-Delaney A12 Effect

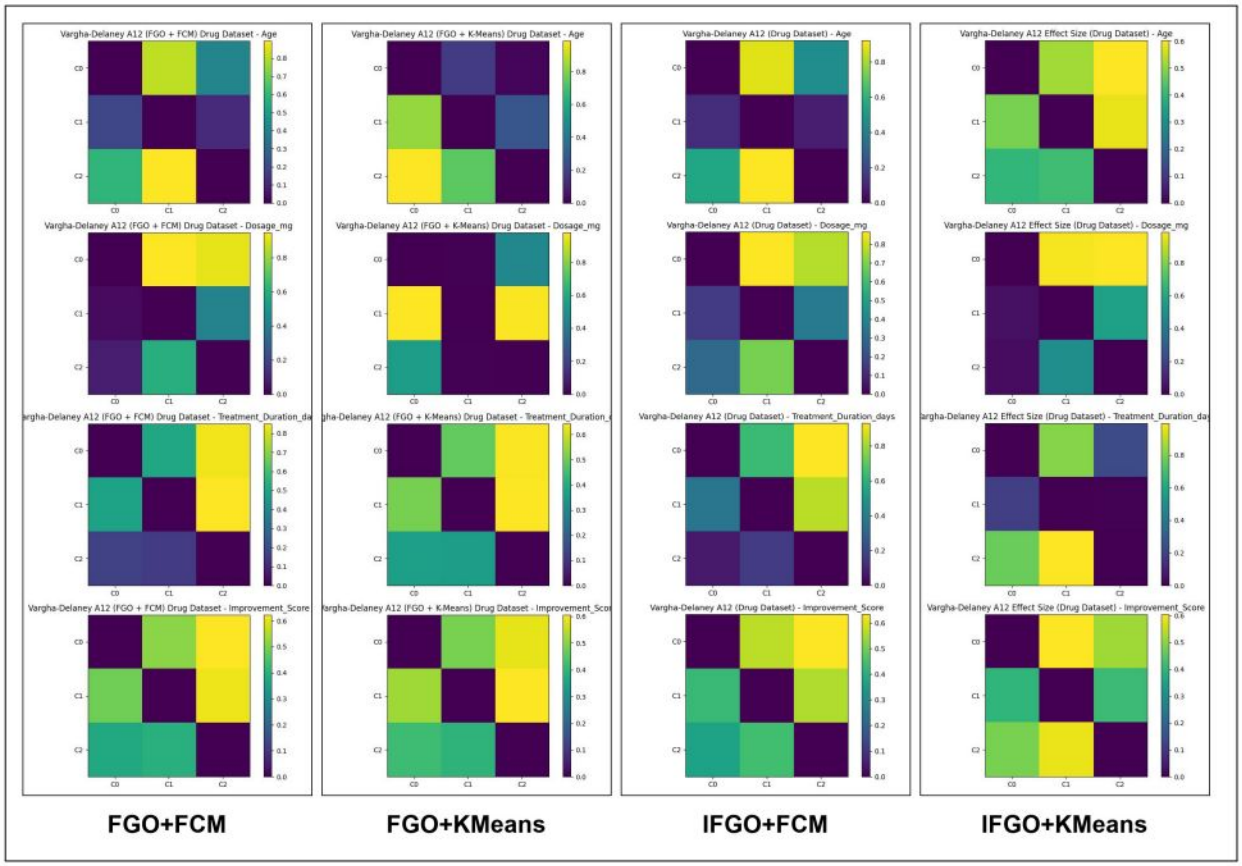


FIGURE 7.9: Vargha–Delaney A12 Effect Size for the Delivery Dataset

Figure 7.9 indicates a stronger distribution shift between clusters, more meaningful structural separation and higher practical significance for IFGO-based variants. Vargha–Delaney adds a larger effect magnitude, stronger cluster dominance and practical robustness. For the Delivery Dataset IFGO variants indicate strong dominance between clusters by providing higher A12 values (0.6–0.9). FGO variants show more moderate values (0.3–0.7) which reflects weaker effect sizes and less pronounced distributional differences.

Chapter 8

Real Life Application

8.1 Aviation Network

8.1.1 Problem Statement

India's domestic aviation network has a large number of airports which are interconnected through complex flight routes. Introducing a better clustering algorithm is a challenging task. Due to uneven geographical distribution along with varying passenger demand, regional connectivity gaps, and operational cost constraints one better algorithms is needed.

8.1.2 Approach Used

K-Means version of IFGO does hard clustering, each airport is assigned to one cluster. IFGO optimizes initial centroid selection to improve clustering stability and performance. IFGO integrated with FCM performs soft clustering. Airports can belong to multiple clusters with varying membership degrees. It enhances centroid positioning for better fuzzy partitioning.

8.1.3 Data Source

TABLE 8.1: Description of the Datasets used in Indian Aviation Network

Dataset	Dimension	Description
City Dataset	4231×4	Contains city-related information.
Flight Dataset	113339×11	Contains flight-related operational data.

Table 8.1 summaries the datasets which are used to demonstrate the aviation network real life application for the proposed algorithm. These datasets are available in the public domain, downloaded from the website Kaggle [24].

8.1.4 Application

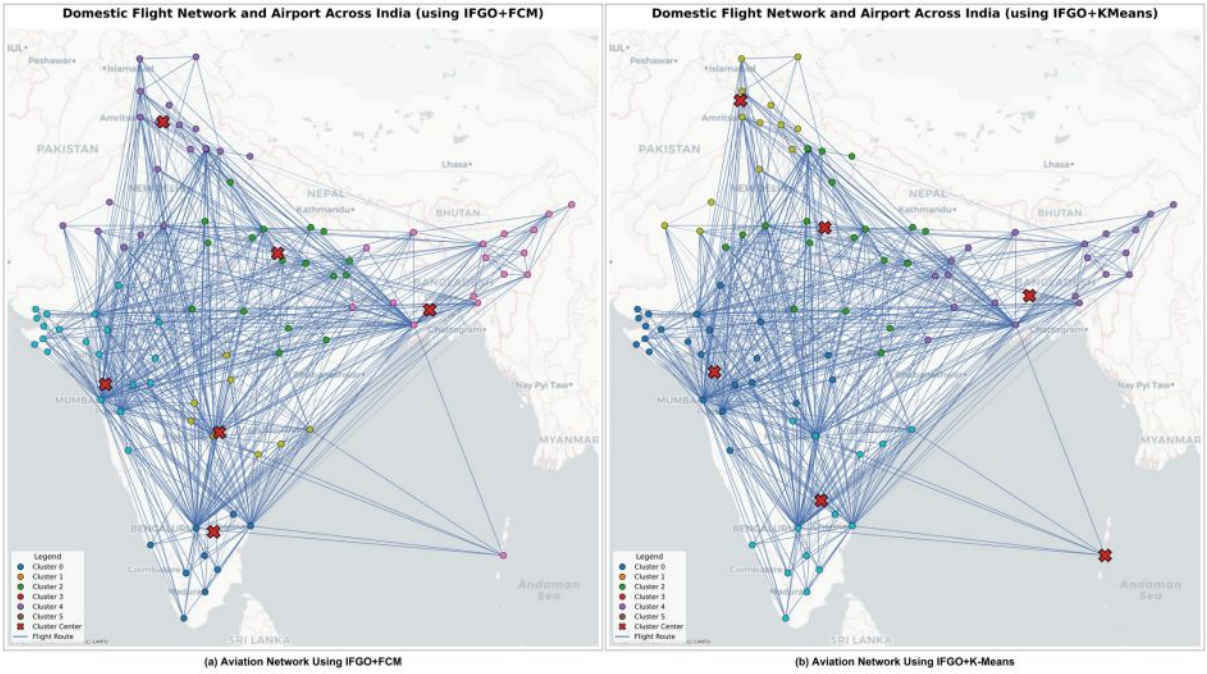


FIGURE 8.1: Implementation on Indian Domestic Flight Dataset

Figure 8.1 displays that IFGO based hybrid algorithms produce clear cluster. Result shows natural regional divisions in domestic flight network along with strategic geographical positioning.

8.1.5 Analysis

West and South India show very dense route connections. It reflects high economic activity, greater airport concentration and strong domestic travel demand. The North-East airports with limited connectivity illustrate regional isolation due to geography. Interconnected routes demonstrates that Indian Domestic aviation is highly dense for which optimized hub and spoke management is needed. IFGO based algorithm effectively determine optimal hub location along with meaningful regional cluster which demonstrates practical applicability for the proposed algorithm.

8.1.6 Insights

IFGO introduces differential exploration and Lévy flight mutation which allows the algorithm to search many possible cluster center positions instead by avoiding the chance of getting stuck. In the aviation map, IFGO provides meaningful formed clusters by placing the centroids in the highly active areas. Traditional optimization algorithms can get stuck in local minima while clustering, on the other hand IFGO helps to escape poor solutions and refine cluster centers. The proposed algorithm shows enhanced exploration, mutation strategy and adaptive search resulting in more structured and meaningful airport network clusters.

8.2 Credit Card Anomaly Detection

8.2.1 Problem Statement

In modern time the usage of digital payments and online banking is increasing, along with it credit card fraud has become a major financial concern. These fraudulent transactions cause financial losses. Often the traditional rule-based algorithms are often unable to detect new sophisticated fraud patterns. There is a need for better algorithm that can automatically identify unusual and suspicious transactions.

8.2.2 Approach Used

The approach typically involves collecting transaction data and pre-processing it to remove noise and irrelevant information. IFGO+FCM provides more refined and adaptive cluster boundaries where, IFGO+KMeans provides better computational efficiency.

8.2.3 Data Source

TABLE 8.2: Description of the Datasets used in Credit Card Anomaly Detection

Dataset	**Dimension**	**Description**
Credit Card Dataset	$284,807 \times 31$	Provides transaction related information.

Table 8.2 summaries the datasets which are used to demonstrate the credit card anomaly detection real life application for the proposed algorithm. These datasets are available in the public domain, downloaded from the website Kaggle [24].

8.2.4 Application

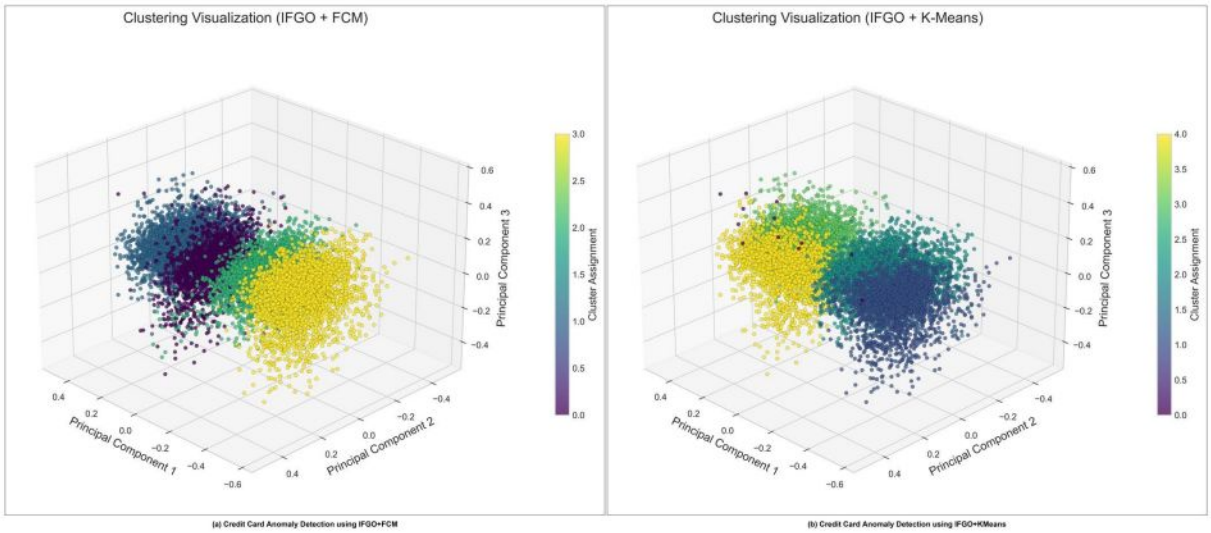


FIGURE 8.2: Implementation for Detecting Unusual Activities in a Credit Card Dataset

Figure 8.2 shows the effectiveness of the IFGO-based hybrid approach to detect the abnormality of the credit card transactions. By leveraging fuzzy membership, IFGO+FCM algorithm forms more refined and adaptive cluster boundaries which enables better separation between the normal and the anomalous transactions comparatively. On the other hand IFGO+KMeans provides better speed of execution for large-scale credit card transaction analysis.

8.2.5 Analysis

For the Credit Card Dataset IFGO successfully optimizes cluster formation in both hybrid methods. FCM provides more flexible clustering, it is more useful for detecting uncertain patterns. Anomalous transactions typically appear near cluster boundaries, far from centroids or as sparsely distributed data points. K-Means based hybrid provides clearer cluster separation which is useful for identifying distinct transaction groups.

8.2.6 Insights

IFGO-based hybrid methods improve clustering performance by optimizing centroid initialization and by preventing algorithms from getting trapped in local minima. These leads to better cluster separation, more stable convergence and improved detection of anomalous transactions. As a result IFGO based hybrid algorithm provides more meaningful output.

Chapter 9

Analysis

9.1 Inspection

The experimental analysis demonstrates that the proposed Improved Fungal Growth Optimization (IFGO) approach provides noticeable improvements in clustering quality, stability and convergence behavior. Across multiple datasets proposed algorithm consistently show stronger performance in terms of cluster validity indices. Furthermore, statistical analyses validate the robustness and statistical significance of the proposed approach. These results confirm that IFGO-based variants often achieve more consistent and statistically distinguishable clustering structures comparatively.

9.2 Advantages

The hybridization of IFGO with K-Means and Fuzzy C-Means (FCM) provides flexibility in clustering. The K-Means variant offers faster computation and clear cluster boundaries, while the FCM variant allows soft clustering. IFGO demonstrates strong robustness across different datasets, including high-dimensional and complex data.

9.3 Limitation

Despite of its improved performance the proposed algorithm has limitations. One of the primary limitations is computational complexity. Since IFGO integrates multiple optimization mechanisms such as chaotic initialization, Lévy flight mutation and opposition-based learning the overall computational cost can become higher compared to traditional clustering methods. Another limitation is the dependence on parameter tuning. The algorithm may face challenges with extremely high dimensional datasets. In such cases, the optimization process may require more iterations to find an optimal solution.

Chapter 10

Conclusion and Future Scope

10.1 Conclusion

Improved Fungal Growth Optimization (IFGO) based hybrid algorithms display superior consistency along with multidimensional robustness. While maintaining variance stability and structural integrity in simpler settings IFGO based algorithms provide complete feature level statistical separability. IFGO exhibits reduced feature bias with improved dispersion control and stronger cross domain reliability. The experimental results confirm the efficaciousness of the proposed algorithm for real-world applications. To improve the computational efficiency for large-scale datasets parallel optimization techniques can be introduced with the proposed algorithm. In addition to improving computational efficiency, the proposed IFGO-based clustering approach demonstrates a strong ability to maintain stability across different datasets. IFGO is capable of preventing premature convergence which ensures that the algorithm continues to search for better cluster configurations throughout the optimization process. This indicates another important aspect of the proposed approach. The obtained results indicates improved clustering accuracy and stability compared with other clustering approaches. Integrating the proposed IFGO algorithm with deep learning techniques may provide new opportunities for improving clustering performance in high dimensional data environments. This could enhance the ability of the algorithm to extract meaningful patterns from large and complex datasets. Overall, the findings of this study highlight the effectiveness of the IFGO based clustering approach as a powerful optimization framework for data clustering problems. The proposed method not only improves clustering quality but also provides a flexible and scalable solution. In summary, by introducing improved search capabilities, enhanced

robustness and better adaptability to different types of datasets with FGO contributes to the advancement of optimization based clustering techniques.

10.2 Future Scope

Clustering has a wide range of applications across various dimains. Enhancing clustering techniques can contribute significant and diverse benefits to society. Some of the key application areas and their impacts are presented in the table below.

TABLE 10.1: Usage of Improved Clustering Algorithms Across Domains

Domain	**Usage**	**Impact**
Healthcare	Grouping patients with similar symptoms or medical history	Enables accurate disease diagnosis, personalized treatment planning, and early detection of potential health risks.
Marketing and Business	Customer segmentation based on purchasing behavior	Supports targeted marketing strategies, improves customer satisfaction, and increases business efficiency.
Social Networks	Community detection and grouping users with similar interests	Enhances recommendation systems and improves online user engagement.
Image Processing	Image segmentation and pattern recognition	Applied in medical imaging, facial recognition, and automated object detection systems.
Cybersecurity	Detecting abnormal patterns in network traffic	Helps identify cyber attacks, fraudulent activities, and potential security threats early.
Bioinformatics	Clustering gene expression data	Helps researchers understand genetic diseases and accelerate drug discovery.
Recommendation Systems	Grouping users with similar preferences	Improves recommendation of products, movies, and online content.

Table 10.1 summarizes the wide range of application of clustering.

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